

Selective Authorization of Automated Performance Patches: Rule Audit and Case Study of a Frozen Promotion Gate

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Abstract

Coding agents can propose performance patches faster than noisy hardware can validate them. This creates an authorization problem distinct from patch generation: when is the evidence strong enough to permit automatic promotion? We audit a frozen promotion gate rather than proposing a post-hoc replacement. Its direct path combines a practical-effect estimand with an uncalibrated directional-consistency threshold. Under an IID Bernoulli-sign interpretation, certifying $q > 0.9$ with a one-sided 95% lower bound requires at least $n = 29$ observations and, at that minimum, every paired difference must be positive; the frozen ladder stops at 4/8/12. Arm order is also perfectly confounded with pair-index parity, so an order effect is not identifiable.

The accompanying preregistered public case study evaluates the gate on fmt 12.1.0 using 30 strict A/A null candidates and one real buffer-reuse candidate. The gate promoted 0 of 30 nulls; under an independent, common-probability Bernoulli model, the corresponding one-sided 95% upper bound is 9.503%. All null candidate and control binaries were byte-identical and shared one hash, so this result calibrates the complete pipeline on strict A/A inputs rather than on semantically equivalent but binary-distinct patches. A deliberately weak first-pair counterfactual would have accepted 16/30 nulls. The real candidate showed a median paired gain of 35.861 ms (6.515%) but ended as `accepted_noisy_single`, an evidence-only verdict requiring independent replication. The report contributes a claim-bounded rule audit, a reproducible case study, and a preregistered design for a later authorization-frontier study; it does not claim a universal false-promotion rate, a superior gate, or cross-system generality.

Scope. The next-stage simulation contract is unexecuted; all related policy comparisons are prospective.

1 Introduction

Performance optimization is increasingly treated as a long-horizon coding agent task. Recent systems learn or generate performance-improving edits across programs and repositories [10]. Generation, however, does not settle whether a proposed patch should be merged. Runtime evidence on shared hardware is affected by process noise, host state, and multilevel variation; rigorous benchmarking therefore requires explicit experimental units, repetition, and uncertainty rather than one favorable timing observation [6, 8]. Automated tuning under unstable measurements faces the same problem [5].

We study a narrower decision: *when should an automated system receive authority to promote a performance patch?* Candidate generation and promotion are different roles. A search procedure may produce many speculative edits, while a promotion authority must fail closed when correctness, practical effect, or measurement quality is unresolved. This distinction is especially important when a statistically interesting result remains too noisy for an unattended merge.

The case study freezes one promotion procedure before timing and evaluates it on one public C++ target. Thirty strict null candidates provide an empirical calibration of end-to-end false promotion behavior. One real buffer-reuse candidate checks that the procedure is not evaluated only on nulls. The design uses paired candidate/control observations, a bootstrap interval, a paired permutation statistic, a minimum practical effect, a directional consistency filter, host-weather gating, and bounded escalation. These are established statistical and measurement components, not individually claimed as methodological inventions [4, 7].

This report makes four claim-bounded contributions:

1. It audits the frozen authorization semantics. The direct path conjoins an effect threshold with directional consistency without a stated joint operating-characteristic calibration, and a $q > 0.9$ confidence interpretation is infeasible within the frozen 4/8/12 ladder under the IID binomial model.

2. It reports a fully replayable public case study: 0 of 30 strict A/A nulls were promoted, while a real candidate was retained as evidence but not automatically promoted. All null candidate and control binaries were byte-identical, so the null result calibrates this end-to-end A/A pipeline rather than behavior on binary-distinct semantic nulls.
3. It shows why binary identity is part of the result rather than an implementation detail. A strict A/A calibration with identical binaries cannot measure sensitivity to layout or alignment changes in semantically equivalent, binary-distinct patches; such studies should report whether the timed artifacts actually differ.
4. It identifies a design limitation: deterministic alternation makes arm order a function of pair-index parity, preventing identification of an arm order effect. Sequence diagnostics are reported only as post-hoc exploratory evidence.

The contribution is consequently not “a new bootstrap gate.” It is an empirical and analytical audit of how a concrete automated promotion policy allocates authority, abstention, and replication. A later, separately frozen study will compare alternative policies across risk, power, abstention, and cost; no such frontier is claimed here.

2 Related Work

Prior work shows that performance evidence can be distorted before any decision rule is applied. Mytkowicz et al. [9] document how seemingly innocuous experimental choices can produce wrong performance data. STABILIZER [3] addresses this class of problem with randomization and a statistically grounded measurement methodology. Duet Benchmarking [1] studies interference-aware measurement in the cloud, where co-running activity can obscure the effect of the program under test. These works motivate the present emphasis on paired observations, host weather, and explicit uncertainty, but they do not specify when a system should be granted authority to promote a code edit.

At a larger operational scale, FBDetect [11] targets tiny performance regressions through in-production monitoring at Meta. Its unit of concern is regression detection across a production fleet; the unit here is a candidate patch entering an automated promotion pipeline under a bounded measurement budget. The distinction is consequential: a detector may identify a likely regression without defining a policy for abstention, independent replication, or automatic authorization. This report therefore treats the preceding systems as measurement and detection neighbors, not as competing promotion gates. Its contribution is to make the authorization semantics of a concrete frozen gate explicit and auditable, while leaving any replacement policy to a separately frozen study.

Relationship to a companion manuscript. The gate audited here is the same frozen `paired_bootstrap_permutation.v1` judge that an earlier manuscript by the same author, unpublished at the time of writing, evaluates on a different subject. That manuscript asks whether such a gate can hold a low promotion rate under noise, and its principal arm uses a constructed candidate set whose candidate-level rows are withheld. This report asks a different question – what the decision rule can and cannot certify – on a separately preregistered study whose subject is `fmt 12.1.0` and whose complete candidate-level evidence is released. The two share a measurement methodology and an author; they do not share a subject program, a measurement run, or any candidate-level evidence.

3 Audited Promotion Procedure

3.1 Authority and outcomes

The procedure separates search from promotion. A candidate first has to pass the frozen semantic correctness contract. It is then timed against a control inside the same locked host window. A positive paired difference $d_i = t_{i,\text{control}} - t_{i,\text{candidate}}$ indicates that the candidate was faster.

The frozen public judge can emit three operational classes relevant here:

- **accepted:** direct automatic promotion;
- **accepted_noisy_single:** statistically positive but noisy evidence, retained for an independent replication and not promoted;
- **NOISY_PENDING** or **rejected:** no promotion authority from the current run.

An independently replicated noisy signal may later become **accepted_noisy_replicated**, which is a promotion verdict. This study did not authorize or execute that replication.

3.2 Frozen statistics and ladder

At a decision depth, the judge computes the median paired improvement $\hat{\theta}$, a two-sided 95% bootstrap interval $[L, U]$, a paired permutation p -value, and the fraction of pair signs agreeing with the sample median. The minimum practical effect is

$$\delta_{\min} = 0.005 \tilde{t}_{\text{control}}. \quad (1)$$

The interval margin, width, and decisiveness are

$$M = L - \delta_{\min}, \quad W = U - L, \quad K = M/W. \quad (2)$$

The direct, *decisive* confidence tier requires

$$M > 0, \quad K \geq 1.0, \quad \hat{q}_{\text{match}} \geq 0.9. \quad (3)$$

A *marginal* tier requires $M > 0$ and permutation $p \leq 0.05$. Marginal evidence on a noisy host can become `accepted_noisy_single`; it is not direct promotion.

The sample ladder is m4, m8, and m12. m4 is screening-only and has no terminal promotion authority. m8 is the first formal decision, while m12 is reached only through the frozen escalation rule. Bootstrap and permutation procedures use 2,000 resamples each. Although the interval and permutation components have nominal settings, the conjunction in Equation 3, the data-dependent ladder, and the replication route were not jointly calibrated to a stated false-authorization probability.

3.3 Counterfactuals

The preregistered *naive-k1* counterfactual accepts whenever the first paired difference is positive. It is intentionally weak and is not presented as a competitive statistical baseline. A separate bootstrap-only sensitivity accepts when the final 95% bootstrap interval has $L > 0$. The latter analysis was defined after the experiment and is therefore labeled post-hoc exploratory. It omits the practical-effect margin, permutation evidence, directional consistency, and decisiveness requirements.

4 Study Design

4.1 Subject and workload

The public subject is fmt 12.1.0 at source commit 407c905e45ad75fc29bf0f9bb7c5c2fd3475976f. The frozen wrapper performs 1,024,000 formatting operations per process. Each observation launches a fresh process of a previously compiled binary. Candidate and control are measured as an ordered pair under one validation lock, and candidates execute sequentially.

The host-weather gate uses five control measurements before candidate timing. The first attempt was classified NOISY and produced no candidate observations. A second attempt was classified QUIET and supplied the only formal candidate window. This preserves the failed preflight without mixing it into the candidate denominator.

4.2 Strict nulls and real candidate

The frozen denominator contains 30 strict null candidates and one real candidate, `fmt_buffer_reuse`. The null transformations passed a pre-timing executable-token equivalence check and, in the compiled formal artifacts, every null candidate binary was byte-identical to its control. All 30 nulls also shared the same SHA-256 value beginning 6e50bccc4189c8aa. The real candidate binary, beginning 52d798dec09f688f, differed from control.

This distinction defines the null estimand. The null experiment evaluates the complete measurement and decision pipeline under A/A binary identity. It does not estimate false promotion for source changes that preserve semantics while altering code layout, alignment, or generated machine code.

The real candidate reuses an output buffer to reduce repeated work. It is a single functional control, not a power sample or a representative population of agent-generated patches. All 31 candidates passed correctness, so this study confirms execution of the correctness gate but does not estimate its ability to reject deliberately incorrect candidates.

4.3 Preregistration, custody, and replay

The target, source, workload, candidate set, ladder, judge, and denominator were frozen before formal timing. The final execution manifest, authorization, and append-only ledger are bound by SHA-256. Registration is self-custodied and hash-bound; it was not deposited with an external registry. The authorized run completed all 31 terminal candidate rows. A later command-line post-processing failure occurred after measurement completion; recovery regenerated only derived summary files and recorded that no measurement row changed.

The public v2 package is a typed, derived representation of the frozen custody. Its self-contained standard-library recomputer reconstructs all 31 terminal judgments and 63 cumulative stage judgments, verifies pair semantics and bindings, and fails closed under pair tampering. It does not import the formal runner. The public judge extraction also matches the original judge on 64 fixed-seed differential cases. This is observed-input and test-matrix equivalence, not proof of full input-domain program equivalence.

4.4 Endpoints and uncertainty

The primary descriptive endpoint is the count of automatic promotions among the 30 strict nulls. For comparison with the frozen report, we also show a one-sided 95% Clopper–Pearson upper bound [2]. Its coverage interpretation requires candidate-level outcomes to behave as independent Bernoulli draws with a common promotion probability. Without that model, the unconditional observation is only the numerator and denominator.

Secondary endpoints are the naive-k1 result, the post-hoc bootstrap-only sensitivity, the real-candidate outcome, and total paired measurement count. No fixed-depth cost comparator was preregistered, so pair-count comparisons to always-m8 or always-m12 are descriptive rather than efficiency estimates.

5 Frozen Results

5.1 Strict-null outcome and its scope

The frozen gate automatically promoted 0 of 30 strict nulls. Eighteen ended as `NOISY_PENDING` and 12 as `rejected`; no candidate was removed from the denominator. Under the candidate-level independent common-probability Bernoulli model, zero events in 30 trials give the one-sided 95% upper bound

$$1 - 0.05^{1/30} = 0.09503, \quad (4)$$

or 9.503%. This is not evidence that the population false-promotion probability is zero.

The binary result must be read beside that numerator: all 30 null candidate/control pairs were byte-identical, and all shared one binary hash. Accordingly, 0/30 is an end-to-end A/A calibration of this implementation. It does not establish the gate’s behavior on binary-distinct semantic nulls.

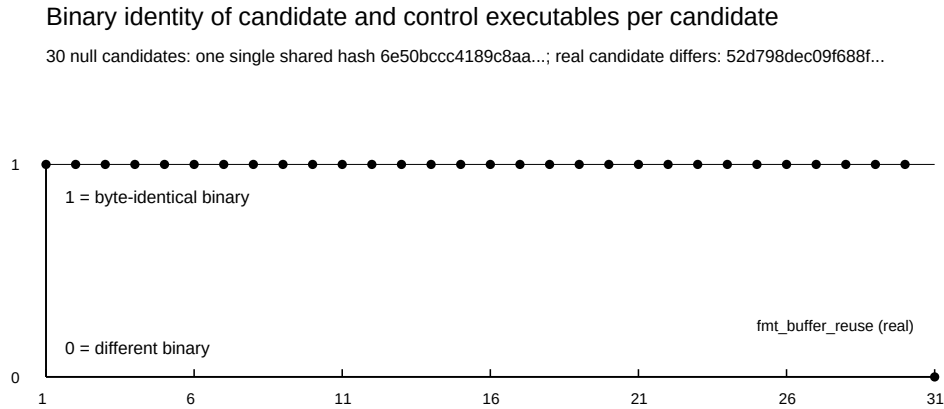


Figure 1: Binary identity in the frozen denominator. All 30 strict null candidate binaries equal control and share one hash; the real candidate is binary-distinct. The figure describes the tested null class rather than a broader semantic-equivalence population.

5.2 Counterfactual decisions

Table 1 compares the frozen result with two explicitly bounded counterfactuals. Naive-k1 would have accepted 16/30 nulls. It also missed the real candidate because that candidate’s first paired difference was -9.387 ms. Bootstrap-only accepted 2/30 nulls, **fmt-011** and **fmt-016**, and accepted the real candidate. The two nulls had interval lower bounds of 2.818 ms and 2.573 ms, but permutation p -values of 0.096952 and 0.112944. These same-data results illustrate different filtering behavior; they are not independent estimates of population error or power.

Table 1: Frozen and counterfactual outcomes. Bootstrap-only is post-hoc exploratory; naive-k1 is a preregistered deliberately weak counterfactual.

Rule	Strict-null accepts	Real-candidate result
Frozen promotion gate	0/30	signal only; no promotion
Naive-k1	16/30	missed
Bootstrap-only	2/30	accepted

5.3 Real-candidate evidence

The real candidate reached m8 with a median paired gain of 35.861 ms. Relative to its run-internal control median of 550.473 ms, this is 6.515%. Its 95% bootstrap interval was [6.368, 58.506] ms, paired permutation $p = 0.030985$, and sign consistency 0.875. The confidence tier was **marginal**; the final verdict was **accepted_noisy_single**. Thus the procedure retained a positive signal but did not grant automatic promotion. No independent replication was authorized or run.

This single candidate demonstrates that the gate is not evaluated only by whether it rejects nulls. It does not estimate recall, establish production readiness, or show that 6.515% generalizes across machines and workloads.

5.4 Sampling and environment

The study used 244 null pairs and 8 real-candidate pairs, for 252 total. Thirty candidates terminated at m8; only **fmt-011** escalated to m12. This was four pairs more than always-m8 and 120 fewer than always-m12, but neither fixed-depth policy was a preregistered comparator, so these differences are not claimed as cost savings.

The rejected preflight had a control CV of 0.1399 and relative range 0.3303; the accepted quiet preflight had CV 0.0116 and relative range 0.0282. The control-only calibration median was 532.686 ms, whereas the real candidate’s run-internal control median was 550.473 ms, a descriptive 3.339% cross-session shift. All reported gain percentages use the run-internal paired control, not the earlier calibration value.

6 Rule Audit

6.1 Two quantities in one direct decision

The interval margin addresses an effect-size quantity: whether the median paired gain exceeds $\delta_{\min}$. The sign filter estimates a different quantity, q_{match} , the probability that a pair has the same sign as the population median. A distribution can have median gain above the practical threshold while fewer than 90% of individual pair differences are positive. Equation 3 therefore embeds a directional-reliability requirement in the direct authorization path.

That requirement may be a legitimate policy estimand, but the frozen rule compares only the point estimate $\hat{q}_{\text{match}}$ with 0.9. It does not use a lower confidence bound for q_{match} , and the public protocol does not state a separate acceptable error probability for certifying that quantity. Conversely, if directional reliability is only diagnostic, it should not be interpreted as evidence about the median-effect estimand.

6.2 What the frozen sample budget can certify

Under IID Bernoulli signs, even an all-positive sequence of length n has the one-sided 95% Clopper–Pearson lower bound $0.05^{1/n}$. Certifying $q > 0.9$ therefore requires

$$0.9^n \leq 0.05, \tag{5}$$

95% one-sided lower bound on q from an all-positive sequence: $0.05^{1/n}$

$\text{sign_min} = 0.9$ requires $n \geq \ln(0.05)/\ln(0.9) = 28.4332 \rightarrow n_{\min} = 29$

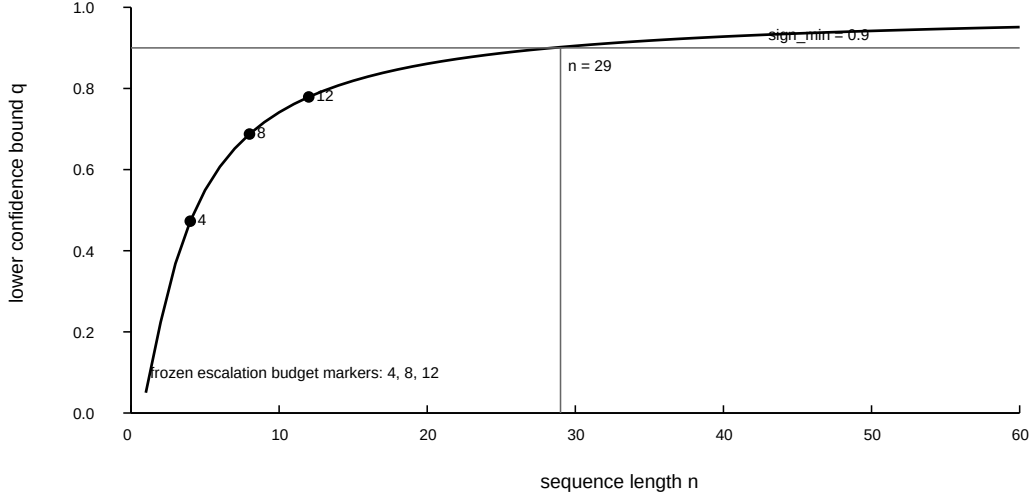


Figure 2: One-sided 95% lower bound after an all-positive IID Bernoulli sequence. The frozen 4/8/12 budget lies below the first sample size that can certify $q > 0.9$.

whose smallest integer solution is $n = 29$. The lower bound is 0.688 at $n = 8$, 0.779 at $n = 12$, and 0.902 at $n = 29$. Thus, if 0.9 is interpreted as a confidence-backed directional-reliability requirement, the frozen m4/m8/m12 budget cannot certify it even when every observed sign agrees. This is a conditional feasibility statement, not an assertion that the existing point filter is itself a 5% test.

6.3 Implicit evidence premium in the decisive tier

For a symmetric interval $[\hat{\theta} - h, \hat{\theta} + h]$, the decisive condition $M/W \geq 1$ implies

$$\hat{\theta} - \delta_{\min} \geq 3h. \quad (6)$$

Using $h \approx 1.96 \text{ SE}$ gives an approximate decisive crossing at 5.88 standard errors, compared with 1.96 standard errors for the interval margin $L > \delta_{\min}$. Under a common $1/\sqrt{n}$ scaling, the corresponding single-stage sample-count ratio is approximately $(5.88/1.96)^2 = 9$.

This calculation is interval geometry, not a complete path-cost comparison. It excludes the permutation condition, sign consistency, discrete ladder, bootstrap-median behavior, and the cost and joint success probability of an independent replication.

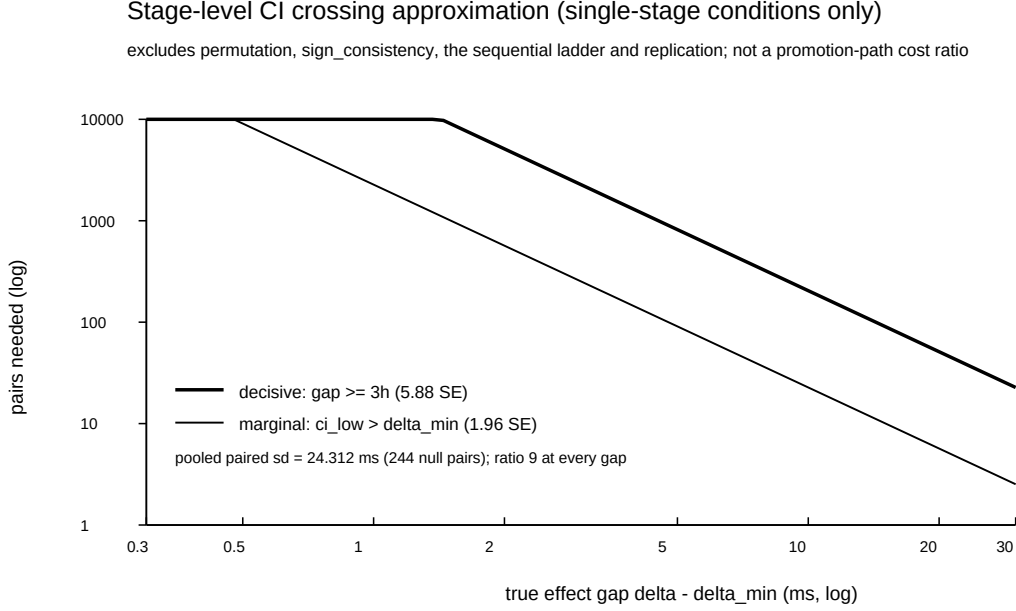


Figure 3: Analytical single-stage CI-crossing approximation using the null-pair scale reported by the deterministic figure generator. The curves are not empirical power and do not represent total direct-versus-replication cost.

6.4 Discrete exact-test reference

As a reference rather than a proposed replacement, an exact one-sided binomial test of $P(d > \delta_{\min}) > 0.5$ has actual size below its nominal 0.05 level because of discreteness. Across every integer n from 8 through 60, the actual IID size ranges from 0.0107 to 0.0494, with mean 0.0347. Reporting power without these changing realized sizes would confound conservatism with policy quality. This reference also answers a different question from the frozen $\hat{q}_{\text{match}} \geq 0.9$ filter.

7 Threats to Validity

Null class and subject scope. The 30 nulls are strict A/A binaries, not binary-distinct semantic nulls. The study has one library, workload, machine, quiet window, and real candidate. Neither the numerator nor the real-candidate result supports cross-project or cross-machine generalization.

Candidate-level probability model. The 9.503% upper bound assumes independent candidate-level Bernoulli outcomes with a common probability. Candidates were executed sequentially on one host, so 0/30 is the primary unconditional fact. The interval is a model-based summary, not a universal guarantee.

Order and dependence. Odd pair indices always ran candidate first and even indices always ran control first: 0 of 244 null pairs violated this mapping. Arm order is thus perfectly confounded with parity, and an arm-order effect is not identifiable. Future paired studies should randomize or counterbalance arm order independently of pair-index parity and record the realized order. Post-hoc sequence diagnostics find more sign alternation than a within-candidate permutation reference, but lag one was selected after the order structure was observed and multiplicity was not controlled. These diagnostics provide exploratory evidence against a naive exchangeability interpretation; they neither identify a mechanism nor prove IID failure.

Rule audit. The $n = 29$ result is exact only under IID Bernoulli signs. The 9-fold result is a symmetric-normal CI-crossing approximation. Neither is a measured risk–power–cost frontier. No alternative rule has been shown superior.

Selection and replication. The real candidate is one selected functional control. Its observed gain is not an unbiased estimate of performance across candidate-generating systems. The required independent replication was not run, so the study cannot report confirmation probability or post-selection shrinkage.

8 Next-Stage Design

A committed draft simulation contract specifies a later comparison of seven policy families across effect, directional reliability, distribution shape, dependence, and budget. It separates the median-effect estimand from directional consistency, includes IID, block-dependent, and period-2 data generating processes, and requires replay, analytical, and differential self-checks. The dependence parameters remain intentionally unfrozen pending the exploratory audit.

The contract is a design artifact, not a result. No Monte Carlo grid has been run, no policy has qualified, and this report makes no recommendation to change the frozen gate. A later version may report a selective-authorization frontier only after the contract is independently reviewed and frozen.

9 Artifact Availability

The public artifact contains typed pair-level evidence, the extracted frozen judge, fixed differential cases, custody and inventory hashes, a self-contained recomputer, and an external-consumer acceptance record. The v0.1 number reproducer independently emits every value used in this report and states the criterion and assumptions for each claim. The pair-dependence audit and all three figures are deterministic under their recorded inputs and seeds.

AI-assisted research provenance. AI coding assistants supported repository implementation, evidence checking, and manuscript drafting. The public package does not retain candidate-by-candidate model attribution or raw agent logs, so this report does not claim that every fnt candidate was agent-generated. Interface model labels, where reported outside the publication package, are dated, author-reported provenance rather than machine-bound experimental factors. The author retained authority over protocol definition and freeze, execution authorization, interpretation, and every claim. Numerical values were accepted only when reproduced by the deterministic public recomputer and the v0.1 number-reproduction script; no model-generated number is treated as evidence.

Private host paths, raw agent logs, lock commands, and untracked evidence are excluded from the publication trees. A recursive scanner verifies the public v1 and v2 packages and reports only redacted finding categories for the private evidence tree.

10 Conclusion

Automatic performance optimization needs an explicit boundary between finding a promising patch and authorizing its promotion. On one frozen public target, the audited gate promoted 0 of 30 strict A/A nulls while retaining one real signal without automatically promoting it. That result is useful but narrow: the null binaries were identical, the candidate-level upper bound is model-dependent, and the single real candidate provides no power estimate.

The rule audit shows why authorization semantics must be stated at the estimand level. A practical median effect, directional consistency, direct evidence, and replicated evidence are different claims with different feasible budgets. The next stage is therefore not to declare the frozen gate correct or incorrect, but to measure how explicit authorization policies trade false promotion, power, abstention, and cost under reviewed dependence models.

References

- [1] Lubomír Bulej, Vojtěch Horký, Petr Tuma, François Farquet, and Aleksandar Prokopec. Duet benchmarking: Improving measurement accuracy in the cloud. In *Proceedings of the ACM/SPEC International Conference on Performance Engineering*, pages 100–107, 2020. doi: 10.1145/3358960.3379132.
- [2] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

- [3] Charlie Curtsinger and Emery D. Berger. STABILIZER: Statistically sound performance evaluation. In *Proceedings of the Eighteenth International Conference on Architectural Support for Programming Languages and Operating Systems*, pages 219–228, 2013. doi: 10.1145/2451116.2451141.
- [4] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993. doi: 10.1201/9780429246593.
- [5] Johannes Freischuetz, Konstantinos Kanellis, Brian Kroth, and Shivaram Venkataraman. TUNA: Tuning unstable and noisy cloud applications. In *European Conference on Computer Systems*, 2025. doi: 10.1145/3689031.3717480. URL <https://arxiv.org/abs/2503.01801>.
- [6] Andy Georges, Dries Buytaert, and Lieven Eeckhout. Statistically rigorous java performance evaluation. In *Proceedings of the 22nd Annual ACM SIGPLAN Conference on Object-Oriented Programming Systems and Applications*, pages 57–76, 2007. doi: 10.1145/1297105.1297033.
- [7] Phillip I. Good. *Permutation, Parametric, and Bootstrap Tests of Hypotheses*. Springer, third edition, 2005. doi: 10.1007/b138696.
- [8] Tomas Kalibera and Richard E. Jones. Rigorous benchmarking in reasonable time. In *Proceedings of the 2013 International Symposium on Memory Management*, pages 63–74, 2013. doi: 10.1145/2464157.2464160.
- [9] Todd Mytkowicz, Amer Diwan, Matthias Hauswirth, and Peter F. Sweeney. Producing wrong data without doing anything obviously wrong! In *Proceedings of the 14th International Conference on Architectural Support for Programming Languages and Operating Systems*, pages 265–276, 2009. doi: 10.1145/1508244.1508275.
- [10] Alexander Shypula, Aman Madaan, Yimeng Zeng, Uri Alon, Jacob Gardner, Milad Hashemi, Graham Neubig, Parthasarathy Ranganathan, Osbert Bastani, and Amir Yazdanbakhsh. Learning performance-improving code edits. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2302.07867>.
- [11] Dong Young Yoon, Yang Wang, Miao Yu, Elvis Huang, Juan Ignacio Jones, Abhinay Kukkadapu, Osman Kocas, Jonathan Wiepert, Kapil Goenka, Sherry Chen, Yanjun Lin, Zhihui Huang, Jocelyn Kong, Michael Chow, and Chunqiang Tang. FBDetect: Catching tiny performance regressions at hyperscale through in-production monitoring. In *Proceedings of the ACM SIGOPS 30th Symposium on Operating Systems Principles*, pages 522–540, 2024. doi: 10.1145/3694715.3695977.